Rational design and modeling of auxetic fiber skin scaffolds for chronic wounds treatment via melt electrowriting

Oscar Lecina-Tejero[1], Jirawarat Iamsamang[2], Miguel Castilho[2], Carlos Borau[1], Elena García-Gareta[1], María Ángeles Pérez[1*]

[1] Multiscale in Mechanical and Biological Engineering, Aragón Institute of Engineering Research (I3A), University of Zaragoza, Spain, angeles@unizar.es

[2] Department of Biomedical Engineering, Eindhoven University of Technology, The Netherlands

Introduction

The development of regenerative biomaterial scaffolds that closely match the mechanical properties of native skin remains a major challenge in tissue engineering, particularly for chronic wound healing. Mechanical metamaterials with auxetic architectures have gained significant attention in recent years, as they enable superior control over mechanical behavior through spatially tailored material designs [1]. However, auxetic scaffold development is still largely empirical, and the complex relationships between microarchitecture, mechanical performance, and manufacturability are poorly understood, which limits their clinical translation for skin regeneration. In this study, our aim was to develop a computational framework for the rational design of auxetic fiber skin scaffolds fabricated by Melt Electro-Writing (MEW) for chronic wound healing. By integrating finite element method (FEM) modeling with experimental validation, we systematically predicted and optimized the mechanical properties of the scaffolds to better match those of native skin.

Methods

The micro-fibrous scaffolds with customizable auxetic re-entrant structures were designed using Python scripts for generating G-code used in fabrication and for model definition in FEM modeling, respectively. Four distinct auxetic structures – HCELL, SREG, SINV, and STRI – were selected for this study to cover the wide range of mechanical properties of skin tissue, and their geometry parameters were adjusted to reduce the pore size while considering printing resolution (Figure 1). FEM modeling was employed to predict the mechanical behavior of the scaffolds. To validate the predictions, the scaffolds were manufactured via MEW and characterized mechanically under biaxial tensile loading.

Results

MEW manufacturing enabled the production of auxetic fiber scaffolds with sub-millimeter pore resolution, while rational design allowed to correct fabrication discrepancies and improve printed design fidelity. Experimental biaxial characterization of the scaffolds revealed design-dependent, biphasic, non-linear mechanical response, and the developed FEM model was capable of not only accurately simulating the observed mechanical response (Figure 2) but also accounting for variations introduced by MEW during fabrication.

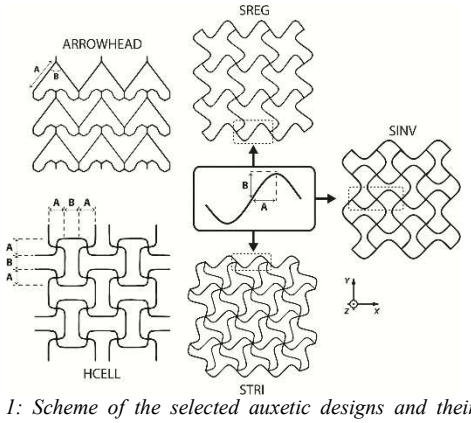


Figure 1: Scheme of the selected auxetic designs and their inner parameters.

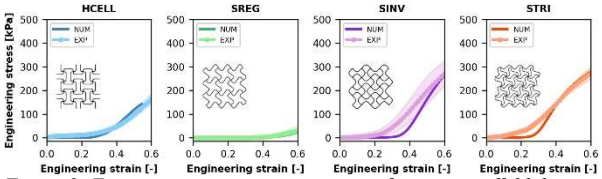


Figure 2: Engineering stress-strain curves of auxetic scaffold designs comparing numerical simulation (NUM) and experimental results (EXP).

Conclusions

The good alignment between numerical predictions and experimental results highlighted the value of FEM modeling as a powerful computational tool for guiding the design and optimization of auxetic scaffolds with tailored mechanical performance for specific soft tissue requirements. The presented methodology offered a significant advancement in the development of functional biomaterials for biomedical applications.

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References

1. Lecina-Tejero et al. J Tissue Eng, 14:1-16, 2023.